DAMIAO | Damiao Technology

DM-J3507-2EC geared motor

User manual V1.0 2024.11.19



Date	Version	Changes
2024.11.19	V1.0	First release

Disclaimer

Thank you for buying the DAMIAO DM-J3507-2EC geared motor (below, the "motor"). Before using this product, read and follow this document and all safety guidance provided by Damiao, otherwise you risk injury to you and those around you, or damage to this product or nearby property. Using this product means you have carefully read this document, and that you understand and accept all terms and content of it and all related documents. You undertake to use this product only for legitimate purposes. You accept full responsibility for using this product and for any consequences. Damiao accepts no liability for damage, injury or any legal consequence arising directly or indirectly from use of this product.

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Cautions

1. 1. Operate the motor strictly within the specified environment and maximum winding temperature, or it will suffer permanent, irreversible damage.
2. 2. Keep debris out of the rotor, or the rotor will run abnormally.
3. 3. Check every part before use. If anything is missing, aged or damaged, stop using it.
4. 4. Ensure correct wiring, and that the motor is mounted correctly and securely.
5. 5. Do not touch the rotor while in use, to avoid accidents. The motor gets hot at high torque output, so take care not to be burned.
6. 6. Do not disassemble the motor yourself. It will affect control accuracy and may cause abnormal operation.

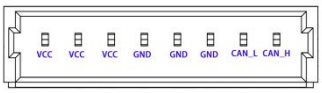
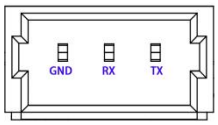
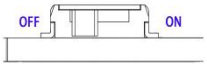
Features

1. 1. Dual encoders give single-turn absolute output-shaft position, retained through power loss.
2. 2. Integrated motor and driver: compact and highly integrated.
3. 3. Supports visual debugging from a host PC, and firmware upgrades.
4. 4. Supports CAN FD, up to 5 Mbps.
5. 5. Speed, position, torque and motor temperature can be reported over the CAN bus.
6. 6. Dual temperature protection.
7. Low speed, high torque.
8. 8. Flexible switching between multiple control modes.

Packing list

Category	Contents
Motor (driver included)	1. Motor (driver included) x1 2. Power + CAN terminal cable: SH1.0 3-pin (200 mm) x1 3. Debug serial signal cable: SH1.0 8-pin (200 mm) x1 An adapter board (SH1.0 3pin+8pin to XT30 +GH1.25) is recommended separately; it is not supplied with the motor.
Motor + USB-to-CAN	1. Motor (driver included) x1 2. Power + CAN terminal cable: SH1.0 3-pin (200 mm) x1 3. Debug serial signal cable: SH1.0 8-pin (200 mm) x1 4. USB-to-CAN debug tool x1 An adapter board (SH1.0 3pin+8pin to XT30 +GH1.25) is recommended separately; it is not supplied with the motor.
Motor + USB-to-CAN + Adapter board	1. Motor (driver included) x1 2. Power + CAN terminal cable: SH1.0 3-pin (200 mm) x1 3. Debug serial signal cable: SH1.0 8-pin (200 mm) x1 4. USB-to-CAN debug tool x1 5. Adapter board (SH1.0 3pin+8pin to XT30 +GH1.25)
Motor + Adapter board	1. Motor (driver included) x1 2. Power + CAN terminal cable: SH1.0 3-pin (200 mm) x1 3. Debug serial signal cable: SH1.0 8-pin (200 mm) x1 4. Adapter board (SH1.0 3pin+8pin to XT30 +GH1.25)

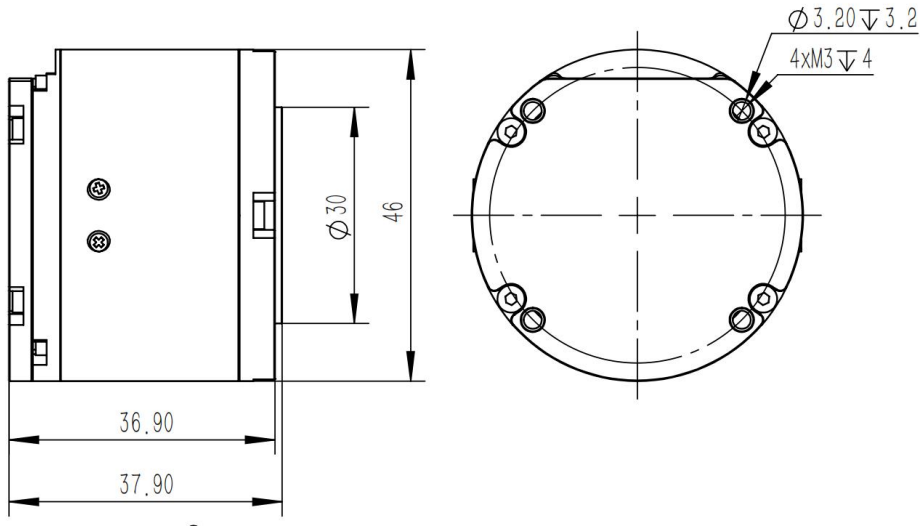
Interfaces and pinouts

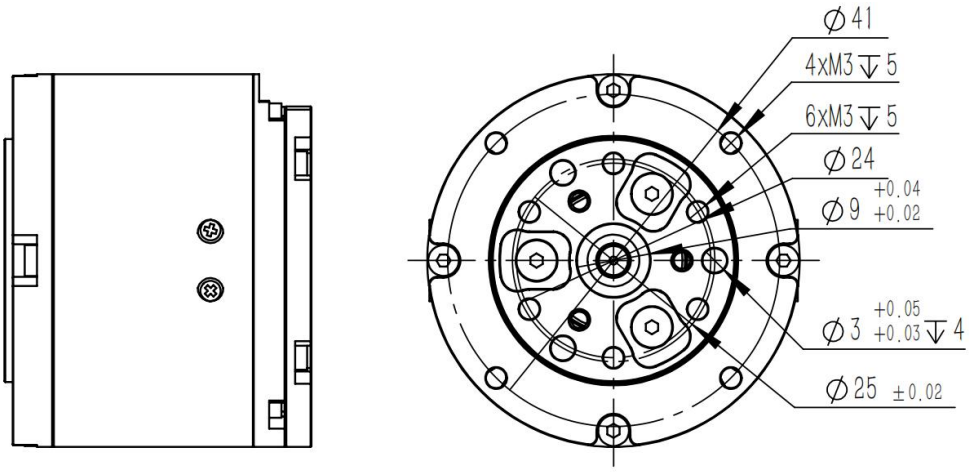
Name - item	Marking	Description
Power + CAN terminal		1. Connect the supply through the SH1.0 8-pin power cable. Rated voltage is 24 V (supports 24-30 V), supplying the motor. 2. Connect external control equipment through equipment, accepting CAN commands and reporting back the motor status.
Debug serial port		Through an SH1.0 3-pin cable, via an Adapter board (SH1.0 3pin+8pin to XT30 +GH1.25) in between, then use the USB-to-CAN debug tool (or a generic USB-to-serial module) to the PC, and use the Damiao tuning assistant to set motor parameters and upgrade firmware.
Termination resistor switch		The motor has a termination resistor, enabled by default.

Note: mind the connector orientation when plugging into the motor port, to avoid skewing or damaging the pins.

Motor dimensions and mounting

Mount the motor to your equipment using the mounting-hole sizes and positions shown.





LED indicator states

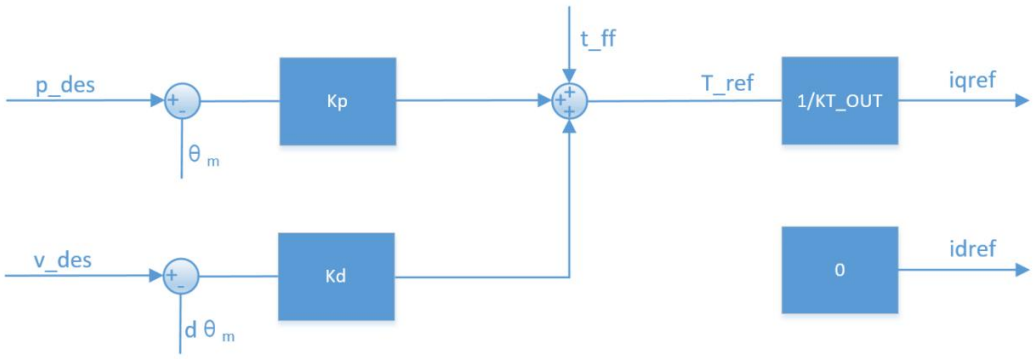
Normal state	Green LED steady	ERR bit 1: enabled mode, normal operation
	Red LED steady	ERR bit 0: disabled mode
Fault state	Red LED flashing	Indicates a fault. The fault types are: 8 ? overvoltage; 9 ? undervoltage; A ? overcurrent; B ? MOSFET over-temperature; C ? motor winding over-temperature; D ? communication lost; E ? overload; Shown in the feedback frame, or on the Damiao tuning assistant screen, to see which fault has occurred.

Operating modes

❖ MIT mode

MIT mode is designed to be compatible with the original MIT mode, allowing seamless switching while flexibly setting the control ranges (P_MAX, V_MAX, T_MAX). The driver converts received CAN data into control variables and computes a torque value as the current-loop setpoint, and the loop drives to that torque current.

The control block diagram is below:



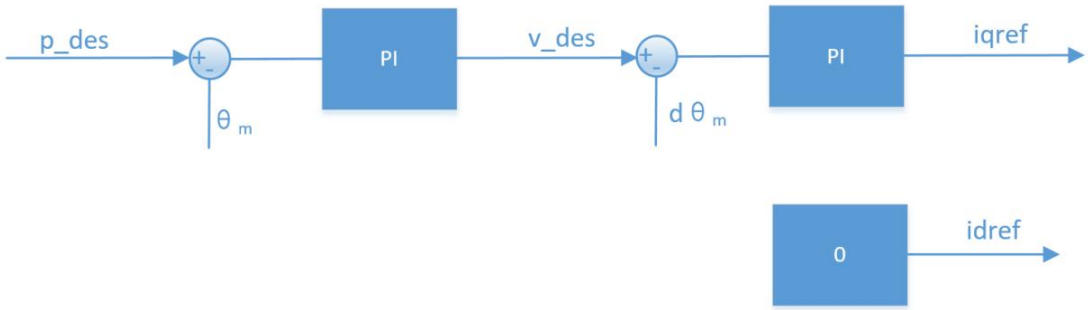
Several control modes derive from MIT mode. With kp=0 and kd non-zero, setting v_des gives constant-speed rotation; with kp=0 and kd=0, setting t_ff gives a commanded torque output.

Note:

1. When controlling position, kd must not be 0, or the motor will oscillate or even run away.
2. In this mode the current-loop bandwidth can be raised to speed up its response.

❖ Position-velocity mode

Position-velocity mode cascades three loops. Position is outermost and its output is the speed setpoint, whose output is the setpoint for the inner current loop, controlling actual current output. See the block diagram the figure below:



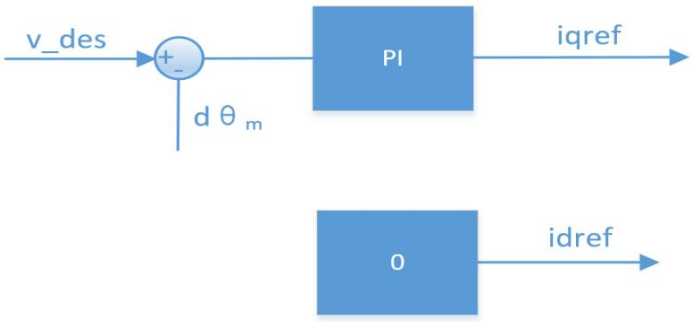
p_des is the target position; v_des limits the maximum absolute speed during the move.

With the gains recommended by the tuning assistant, cascaded position mode reaches good accuracy and a relatively smooth response, though the response time is longer. Besides v_des, acceleration and deceleration can be set. If extra oscillation appears during a move, raise the acceleration/deceleration.

Note: p_des and v_des are in rad and rad/s respectively, of type float.

❖ Velocity mode

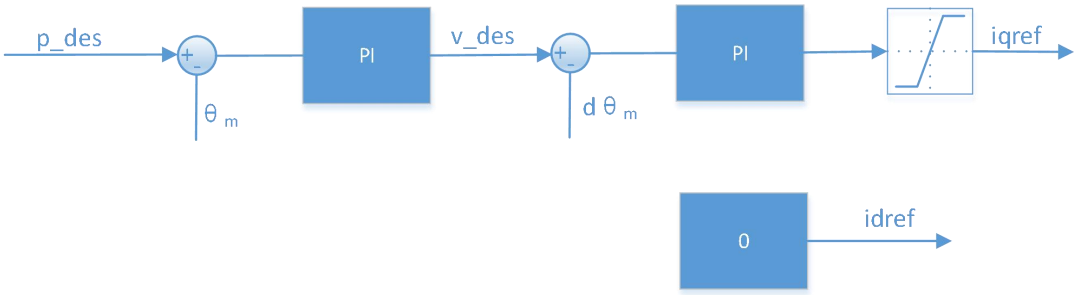
Velocity mode holds the motor at a set speed. The control block diagram is below:



Note: v_des is in rad/s, of type float.

❖ Force-position hybrid mode

Force-position hybrid mode adds dynamic control of output torque on top of position-velocity mode. Its block diagram is below:



A current saturation stage follows the speed-loop output, so the current-loop setpoint stays within the given range.

Control protocol

Control uses the CAN standard frame format at a default 1 Mbps. Commands can switch to other baud rates; see the baud rate section. Frames divide by function into receive and feedback frames. The receive frame is incoming control data used to command the motor, and to read and write controller parameters. Beyond basic configuration, mode and baud rate changes have been added. The feedback frame carries motor status to the host, and feedback is polled: whenever a received frame ID matches the motor's CAN ID (low 8 bits checked, high 3 bits ignored), the driver puts its current status on the bus. Depending on the selected mode, the receive frame format and frame ID differ per mode, but the feedback frame format and data are the same in all modes.

❖ Feedback frame

The feedback frame ID is set in the tuning assistant (Master ID), default 0. It mainly reports motor position, speed and torque. Its frame format is:

Feedback message	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
MST_ID	ID|ERR<<4	POS[15:8]	POS[7:0]	VEL[11:4]	VEL[3:0]|T[11:8]	T[7:0]	T_MOS	T_Rotor

Where:

ID is the controller ID, the low 8 bits of CAN_ID

ERR is the status. The status types are:

0 ? disabled;

1 ? enabled;

8 ? overvoltage;

9 ? undervoltage;

A ? overcurrent;

B ? MOSFET over-temperature;

C ? motor winding over-temperature;

D ? communication lost;

E ? overload;

POS is the motor position*.

VEL is the motor speed*.

T is the motor torque*.

T_MOS is the average MOSFET temperature on the driver, in °C

T_Rotor is the average internal winding temperature, in °C

Position, speed and torque are linearly mapped from float to signed fixed-point, position using 16 bits, and speed and torque 12 bits each.

*Note:

1. After power-on the reported motor position is limited to [-pi, pi] rad.

2. Position is in rad and refers to the output shaft, i.e. after reduction. Descriptions of position descriptions of it follow this definition and are not repeated.

3. Speed is in rad/s and refers to the output shaft, i.e. after reduction. Descriptions of speed descriptions of it follow this definition and are not repeated.

4. Torque is in Nm and refers to the output shaft, i.e. after reduction. Descriptions of torque this definition, and it is not repeated below.

❖ Control frame in MIT mode

Control message	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
ID	p_des [15:8]	p_des [7:0]	v_des [11:4]	v_des[3:0]| Kp[11:8]	Kp [7:0]	Kd [11:4]	Kd[3:0]| t_ff[11:8]	t_ff[7:0]

Frame ID equals the configured CAN ID

P_des: position setpoint

V_des: speed setpoint

Kp: position proportional gain

Kd: position derivative gain

T_ff: torque feedforward setpoint

Each parameter follows the mapping of the previous section. The ranges of p_des, v_des and t_ff are set in the tuning assistant, Kp ranges over [0,500] and Kd over [0,5].

A standard CAN frame carries only 8 bytes, so the MIT command format packs Position, Velocity, Kp, Kd and Torque bitwise into 8 bytes. Position takes 2 bytes / 16 bits, Velocity takes 12 bits, Kp 12 bits and Kd 12 bits.

❖ Control frame in position-velocity mode

Control message	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x100+ID	p_des				v_des			

Frame ID is the configured CAN ID plus an offset of 0x100

P_des: position setpoint, float, low byte first, high byte last

V_des: speed setpoint, float, low byte first, high byte last

The command is sent with CAN ID 0x100+ID. The speed setpoint is the maximum speed used to reach the commanded position, i.e. the speed of the constant-velocity segment.

❖ Control frame in velocity mode

Control message	D[0]	D[1]	D[2]	D[3]
0x200+ID	v_des			

Frame ID is the configured CAN ID plus an offset of 0x200

V_des: speed setpoint, float, low byte first, high byte last

The command is sent with CAN ID 0x200+ID.

❖ Control frame in force-position hybrid mode

The control message is:

Control message	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x300+ID	p_des				v_des		i_des	

P_des: position setpoint in rad, float, low byte first, high byte last;

V_des: speed limit in rad/s, scaled 100x, unsigned 16-bit, low byte first, high byte last, range 0-10000, clamped at 10000, so the actual speed limit is 0-100 rad/s;

I_des: torque-current limit, per-unit, scaled 10000x, unsigned 16-bit, low byte first, high byte last, range 0-10000, clamped at 10000, giving an actual per-unit current limit of 0-1.0

Per-unit current: actual current divided by maximum current. The DM-J3507-2EC maximum is 10.26 A.

❖ Reading parameters

Message ID	Type	D[0]	D[1]	D[2]	D[3]
0x7FF	STD	CANID_L	CANID_H	0x33	RID

RID is the register address; see the appendix <Register list and ranges>

On a successful read the register data is returned, in this frame format:

Message ID	Type	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
MST_ID	STD	CANID_L	CANID_H	0x33	RID	Data			

Data is float or unsigned integer, 32 bits across 4 bytes, low byte D4 and high byte D7. The same applies below.

❖ Writing parameters

Message ID	Type	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x7FF	STD	CANID_L	CANID_H	0x55	RID	Data			

RID as above. On a successful write the data written is returned, in the same frame format as sent.

Message ID	Type	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
MST_ID	STD	CANID_L	CANID_H	0x55	RID	Data			

Written register data takes effect immediately but is not stored and is lost on power-down. Send the save-parameters command to write all modified parameters into the chip.

❖ Saving parameters

Message ID	Type	D[0]	D[1]	D[2]	D[3]
0x7FF	STD	CANID_L	CANID_H	0xAA	0x01

On a successful write the reply format is:

Message ID	Type	D[0]	D[1]	D[2]	D[3]
MST_ID	STD	CANID_L	CANID_H	0xAA	0x01

Note:

1. Saving parameters only takes effect in disabled mode.
2. Saving parameters stores all parameters at once.
3. This writes parameters to on-chip flash, taking up to 30 ms per operation; allow enough time.

4. Flash endurance is about 10,000 cycles; do not send "save parameters" frequently.

❖ Switching modes

Several modes can be switched between. The modes currently supported are:

Code	Mode
1	MIT
2	Position-velocity
3	Speed
4	Force-position hybrid

Change mode by writing the mode register (0x0A). On switching, the motor first zeroes its command values, including position, speed, and the torque feedforward and KP, KD values used in MIT mode.

When switching from another mode into a position mode, read the exact position first to avoid a jolt (register 0x50) before switching, and switch at zero motor speed where possible.

A mode change is not written to flash and is lost on power-down. On the next power-up the control mode is the mode last saved to flash.

❖ Changing the CAN baud rate

Writing specific data to the baud rate register (address 0x23) changes the CAN baud rate.

Specific baud rates can be set. The rates currently supported are:

Code	Baud rate
0	125K
1	200K
2	250K
3	500K
4	1M
5	2M
6	2.5M
7	3.2M
8	4M
9	5M

After a successful change the driver first replies at the old baud rate, then uses the new one. On power-up the motor checks the stored baud rate; above 5 Mbps it defaults to 1 Mbps. Above 1 Mbps (not including 1 Mbps) it switches to CAN FD; at or below 1 Mbps it becomes CAN 2.0B. A motor set to CAN FD can still receive CAN 2.0B data frames, but sends its feedback in CAN FD, so

the host will receive no feedback and the driver will report errors continuously. If a CAN 2.0B host sets the wrong the ID, the baud rate can still be changed back with the baud rate command.

Using the Damiao tuning assistant

Connect PC and motor with the Damiao USB-to-CAN debug tool, and use the Damiao assistant to set parameters and upgrade firmware.

Connect the motor's debug serial port to the PC with a GH1.25 3-pin cable, and the CAN terminal with a GH1.25 2-pin cable to the USB-to-CAN debug tool. Set parameters and upgrade firmware from the Damiao tuning assistant.

With the serial, CAN and power connections made, open the Damiao tuning assistant on the PC, select the serial device and open the port. Power the motor and the port prints information; Control Mode shows the current drive mode.

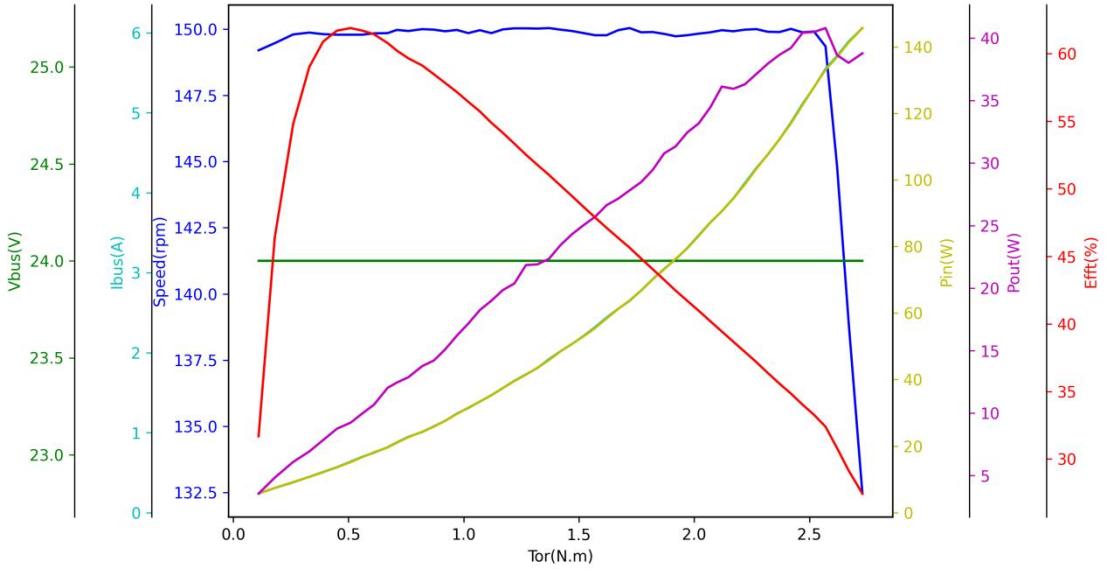
For the detailed procedure see: Tuning Assistant Manual (Damiao Drive Control Protocol) V1.4.pdf

Download link:

https://gitee.com/kit-miao/damiao/tree/master/%E5%85%B3%E8%8A%82%E7%94%B5%E6%9C%BA/%E6%8E%A7%E5%88%B6%E5%8D%8F%E8%AE%AE

Characteristic curves

Test conditions: 25 °C room temperature, laboratory environment.



Characteristic parameters

Use the motor within the parameters below.

Motor parameters	Rated voltage	24V
	Rated phase (supply) current	3.0A(1.2A)
	Peak phase (supply) current	8.3A(4A)
	Rated torque	0.8Nm
	Peak torque	3Nm
	Rated speed	150rpm
	Maximum no-load speed	460rpm
Motor characteristics	Reduction ratio	7: 1
	Pole pairs	14
	Phase inductance	235uH
	Phase resistance	0.8Ω
Structure and mass	Outer diameter	46mm
	Height	37.9mm
	Motor mass	approx. 150 g
Encoder	Encoder resolution	14-bit
	Number of encoders	2
	Encoder type	Magnetic, single-turn (absolute over one output-shaft revolution)
Communication	Control interface	CAN
	Tuning interface	UART@921600bps
Control and protection	Control mode	MIT mode
		Velocity mode
		Position mode
		Force-position hybrid mode
	Protection	Driver over-temperature protection at 120 °C; on over-temperature the motor exits "enabled mode"
		Motor over-temperature protection, set as required; recommended not above 100 °C; on over-temperature the motor exits "enabled mode"
		Motor overvoltage protection, set as required; recommended not above 30 V; on overvoltage it exits "enabled mode"
		Communication-loss protection: if no CAN command arrives within the set period it automatically exits "enabled mode"
		Motor overcurrent protection, set as required; recommended not above 9.8 A; on overcurrent it exits "enabled mode"
		Motor undervoltage protection: if the supply falls below the set value it exits "enabled mode". Supply voltage must not fall below 15 V

Appendix: register list and ranges

Address (HEX)	Address (DEC)	Variable	Description	R/W	Range	Type
0x00	0	UV_Value	Undervoltage protection threshold	RW	(10.0,fmax]	float
0x01	1	KT_Value	Torque constant	RW	[0.0,fmax]	float
0x02	2	OT_Value	Over-temperature threshold	RW	[80.0,200)	float
0x03	3	OC_Value	Overcurrent threshold	RW	(0.0,1.0)	float
0x04	4	ACC	Acceleration	RW	(0.0,fmax)	float
0x05	5	DEC	Deceleration	RW	[-fmax,0.0)	float
0x06	6	MAX_SPD	Maximum speed	RW	(0.0,fmax]	float
0x07	7	MST_ID	Feedback ID	RW	[0,0x7FF]	uint32
0x0A	8	ESC_ID	Receive ID	RW	[0,0x7FF]	uint32
0x09	9	TIMEOUT	Timeout alarm period	RW	[0,2^32-1]	uint32
0x0A	10	CTRL_MODE	Control mode	RW	[0,4]	uint32
0x0B	11	Damp	Motor viscous coefficient	RO	/	float
0x0C	12	Inertia	Motor inertia	RO	/	float
0x0D	13	hw_ver	Reserved	RO	/	uint32
0x0E	14	sw_ver	Software version	RO	/	uint32
0x0F	15	SN	Reserved	RO	/	uint32
0x10	16	NPP	Motor pole pairs	RO	/	uint32
0x11	17	Rs	Motor phase resistance	RO	/	float
0x12	18	Ls	Motor phase inductance	RO	/	float
0x13	19	Flux	Motor flux linkage	RO	/	float
0x14	20	Gr	Gear reduction ratio	RO	/	float
0x15	21	PMAX	Position mapping range	RW	(0.0,fmax]	float
0x16	22	VMAX	Speed mapping range	RW	(0.0,fmax]	float
0x17	23	TMAX	Torque mapping range	RW	(0.0,fmax]	float
0x18	24	I_BW	Current loop bandwidth	RW	[100.0,1.0e4]	float
0x19	25	KP_ASR	Speed loop Kp	RW	[0.0,fmax]	float
0x1A	26	KI_ASR	Speed loop Ki	RW	[0.0,fmax]	float
0x1B	27	KP_APR	Position loop Kp	RW	[0.0,fmax]	float
0x1C	28	KI_APR	Position loop Ki	RW	[0.0,fmax]	float
0x1D	29	OV_Value	Overvoltage threshold	RW	TBD	float
0x1E	30	GREF	Gear torque efficiency	RW	(0.0,1.0]	float
0x1F	31	Deta	Speed loop damping coefficient	RW	[1.0,30.0]	float
0x20	32	V_BW	Speed loop filter bandwidth	RW	(0.0,500.0)	float
0x21	33	IQ_c1	Current loop boost coefficient	RW	[100.0, 1.0e4]	float
0x22	34	VL_c1	Speed loop boost coefficient	RW	(0.0,1.0e4]	float
0x23	35	can_br	CAN baud rate code	RW	[0,4]	uint32
0x24	36	sub_ver	Sub-version number	RO	/	uint32
0x32	50	u_off	u-phase offset	RO	/	float

0x33	51	v_off	v-phase offset	R0	/	float
0x34	52	k1	Compensation factor 1	R0	/	float
0x35	53	k2	Compensation factor 2	R0	/	float
0x36	54	m_off	Angle offset	R0	/	float
0x37	55	dir	Direction	R0	/	float
0x50	80	p_m	Current motor position	R0	/	float
0x51	81	xout	Output shaft position	R0	/	float

RW: read/write.

RO: read only.